

Elevate Labs SQL Developer Internship

Task 4 -Aggregate Functions and Grouping.

- **Aggregate Functions and Grouping –**

- 1. SUM –**

```
CREATE TABLE sales (  
  id INTEGER PRIMARY KEY,  
  sales_date DATE,  
  region VARCHAR(50),  
  product VARCHAR(50),  
  quantity INTEGER,  
  unit_price DECIMAL(10,2)  
);
```

```
INSERT INTO sales (sales_date, region, product, quantity, unit_price)  
VALUES  
( '2025-09-01','North','WidgetA', 10, 20.00),  
( '2025-09-01','North','WidgetB', 5, 50.00),  
( '2025-09-01','South','WidgetA', 8, 20.00),  
( '2025-09-02','North','WidgetA', 12, 20.00),  
( '2025-09-02','South','WidgetB', 7, 50.00),  
( '2025-09-02','East','WidgetC', 3, 100.00),  
( '2025-09-03','East','WidgetA', 4, 20.00);
```

- **Aggregate Function –**

- 1. Total quantity sold (over all sales):**

```
SELECT SUM(quantity) AS total_quantity_sold  
FROM sales;
```

- 2. Total revenue (sum of quantity × unit_price) across all rows:**

```
SELECT SUM(quantity * unit_price) AS total_revenue  
FROM sales;
```

3. Average unit price across all sales:

```
SELECT AVG(unit_price) AS avg_unit_price  
FROM sales;
```

- **Use GROUP BY to categorize (aggregate per category)**

1. Total quantity sold per region:

```
SELECT region, SUM(quantity) AS total_qty  
FROM sales  
GROUP BY region;
```

2. Total revenue per region:

```
SELECT region,  
       SUM(quantity * unit_price) AS total_revenue  
FROM sales  
GROUP BY region;
```

3. Average quantity sold per product:

```
SELECT product, AVG(quantity) AS avg_qty  
FROM sales  
GROUP BY product;
```

4. Count of sales transactions per region:

```
SELECT region, COUNT(*) AS num_sales  
FROM sales  
GROUP BY region;
```

- **Filter groups using HAVING –**

- 1. Regions whose total quantity sold > 15:**

```
SELECT region, SUM(quantity) AS total_qty
FROM sales
GROUP BY region
HAVING SUM(quantity) > 15;
```

- 2. Regions whose total revenue exceeds, say, 1,000:**

```
SELECT region, SUM(quantity * unit_price) AS revenue
FROM sales
GROUP BY region
HAVING SUM(quantity * unit_price) > 1000;
```

- 3. Products whose average quantity sold is ≥ 8 :**

```
SELECT product, AVG(quantity) AS avg_qty
FROM sales
GROUP BY product
HAVING AVG(quantity) >= 8;
```

- 4. Region-product combinations with revenue > 500:**

```
SELECT region, product,
       SUM(quantity) AS sum_qty,
       SUM(quantity * unit_price) AS revenue
FROM sales
GROUP BY region, product
HAVING SUM(quantity * unit_price) > 500;
```